

Day 11 Student Handout

Geometry Summer Credit | Pythagorean Theorem, Converse, Distance Formula, and Radicals

Name:	Date:	Period:
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Learning Targets

I can find missing sides in right triangles. I can use the converse of the Pythagorean theorem to test whether a triangle is right. I can connect the distance formula to right triangles on the coordinate plane.

Lesson 11A: Pythagorean Theorem and Converse

Warm-Up

1.	Simplify: $\sqrt{50}$.
2.	Solve: $x^2 = 144$.
3.	Find the missing value: $\frac{5}{8} = \frac{x}{32}$.

Concept Launch

Right-Triangle Relationship

For a right triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse. The hypotenuse is always across from the right angle and is the longest side.

Problem	Work / notes
The legs of a right triangle are 9 and 12. Find the hypotenuse.	Equation: Answer: Check:
A right triangle has hypotenuse 13 and one leg 5. Find the missing leg.	Equation: Answer: Check:
Do side lengths 8, 15, and 17 form a right triangle?	Largest side: Compare squares: Conclusion:

Converse and Classification

Converse Test

To test whether three side lengths form a right triangle, put the largest side in the c position. If $a^2 + b^2 = c^2$, the triangle is right. If not, it is not right.

4.	Use the converse to test 6, 10, and 13.
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5.	Use the converse to test 10, 24, and 26.
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6.	Use the converse to test 7, 9, and 12.
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Lesson 11B | Distance Formula and Radical Simplification

Lesson 11B: Distance Formula

Coordinate Connection

The distance between two points is the length of the hypotenuse of a right triangle. Horizontal change and vertical change act like the legs.

Problem	Work / notes
Find the distance between A(1, 2) and B(7, 10).	Delta x: Delta y: $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ Answer:
Find the distance between C(-3, 4) and D(5, 7).	Delta x: Delta y: d = Simplified answer:
Simplify $\sqrt{72}$.	Factor out a perfect square: Answer:
Simplify $\sqrt{98}$.	Factor out a perfect square: Answer:

Practice

7.	Find the missing side: legs 7 and 24.
8.	Find the missing leg: hypotenuse 25 and one leg 15.
9.	Find the distance between (-2, 3) and (4, 11).
10.	Find the distance between (0, -5) and (6, 4). Leave your answer in simplest radical form.